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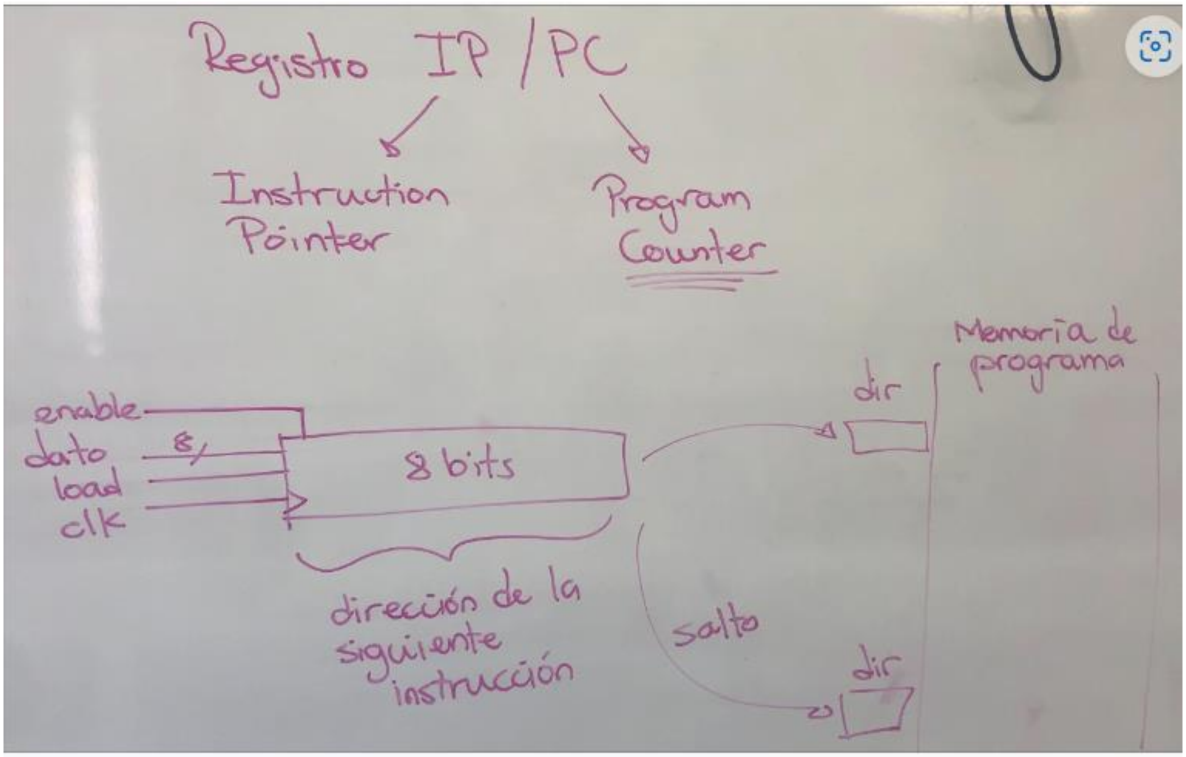
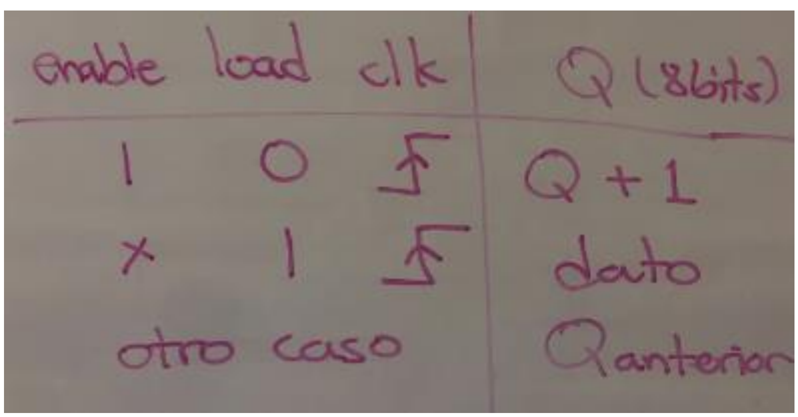
Practice #6: Counters with Verilog in Cadence

Design a program counter or instruction pointer with Verilog in Cadence, based on the next diagram.



The circuit functionality is represented in the next table:



enable	load	clk	Q (8bits)
1	0	↑	Q + 1
x	1	↑	dato
otro caso			Q anterior

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HDL code

```verilog
//Verilog HDL for "Verilog", "Counter" "functional"


module Counter (Enable,Data,Load,Clk,Reset,qout);

input Enable;
input [7:0] Data;
input Load;
input Clk;
input Reset;
output [7:0] qout;
reg [7:0] q;
reg [7:0] Out_s;

always @(posedge Clk) begin
    if(Load == 1'b0 && Enable == 1'b1)begin
        q <= q+1;
    end
    else if (Load == 1'b1)begin
        q <= Data;
    end
    else if(Reset)begin
        q <= 8'b0;
    end
end

assign qout = q;

endmodule

```

Test bench codes

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```
//Verilog HDL for "Verilog", "Counter_tb" "functional"


module Counter_tb ();

reg Enable;
reg [7:0] Data;
reg Load;
reg Clk;
reg Reset;
wire [7:0] qout;

Counter uut(Enable,Data,Load,Clk,Reset,qout);

initial begin

    $display("Start");

    Clk = 1;
    Enable = 0;
    Reset = 0;
    Load = 1;
    Data = 4'b0000; //D
    #10;
    Clk = 0;


    #10;
    Clk = 1;

    #10;
    Clk = 0;

    Reset = 1;
    Clk = 0;
    Data = 4'b1101; //D
    Enable = 1;
    Load = 0;
    Reset = 0;
    #10;
```

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```
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;

    #10;
    Clk = 0;
    Data = 4'b0011; //3
    Enable = 1;
    Load = 0;
    Reset = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;

   #10;
    Clk = 0;
    Enable = 0;
    Load = 1;
    Reset = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;

    #10;
    Clk = 0;
    Enable = 1;
    Load = 0;
    Reset = 0;
    #10;
```

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```
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;

    #10;
    Clk = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;
        #10;
    Clk = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;

    Enable = 0;
    Load = 0;

    #10;
    Clk = 0;
    Data = 4'b1111; //F
    Enable = 1;
    Load = 0;
    Reset = 0;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
```

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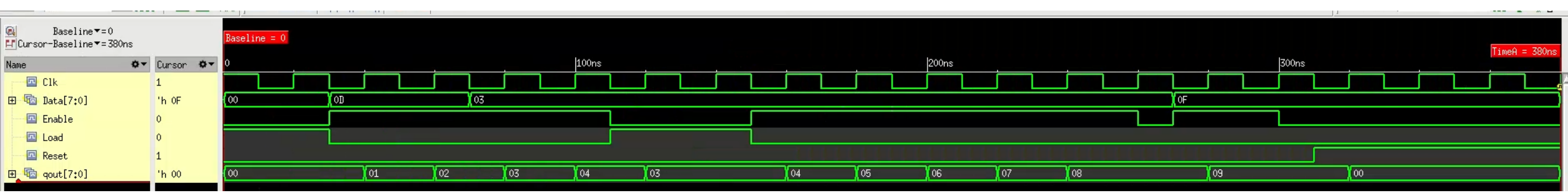
```
    #10;
    Clk = 1;
    Enable = 0;
    Load = 0;
    #10;
    Clk = 0;
    Enable = 0;
    Load = 0;
    Reset = 1;
    #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;

 #10;
    Clk = 0;
     #10;
    Clk = 1;
    #10;
    Clk = 0;
    #10;
    Clk = 1;
    $display("Test completed");

end

endmodule

```

Simulation



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```
module Counter (Enable,Data,Load,Clk,Reset,qout);

input Enable;
input [7:0] Data;
input Load;
input Clk;
input Reset;
output [7:0] qout;
reg [7:0] q;
reg [7:0] Out_s;

always @(posedge Clk) begin
   if(Load == 1'b0 && Enable == 1'b1)begin
      q <= q+1;
   end
   else if (Load == 1'b1)begin
      q <= Data;
   end
   else if(Reset)begin
      q <= 8'b0;
   end
end

assign qout = q;

endmodule
/////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
module Counter_tb ();

reg Enable;
reg [7:0] Data;
reg Load;
reg Clk;
reg Reset;
wire [7:0] qout;

Counter uut(Enable,Data,Load,Clk,Reset,qout);

initial begin


   Clk = 1;
   Enable = 0;
```

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```
Reset = 0;
Load = 1;
Data = 4'b0000; //D
#10;
Clk = 0;


#10;
Clk = 1;

#10;
Clk = 0;

Reset = 1;
Clk = 0;
Data = 4'b1101; //D
Enable = 1;
Load = 0;
Reset = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;
Clk = 1;

#10;
Clk = 0;
Data = 4'b0011; //3
Enable = 1;
Load = 0;
Reset = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;
Clk = 1;

#10;
Clk = 0;
Enable = 0;
Load = 1;
Reset = 0;
```

```
#10;
Clk = 1;
#10;
Clk = 0;
#10;
Clk = 1;

#10;
Clk = 0;
Enable = 1;
Load = 0;
Reset = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;
Clk = 1;

#10;
Clk = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;
Clk = 1;
  #10;
Clk = 0;
#10;
Clk = 1;
#10;
Clk = 0;
#10;
Clk = 1;

Enable = 0;
Load = 0;

#10;
Clk = 0;
Data = 4'b1111; //F
Enable = 1;
Load = 0;
```

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```
  Reset = 0;
  #10;
  Clk = 1;
  #10;
  Clk = 0;
  #10;
  Clk = 1;
  Enable = 0;
  Load = 0;
  #10;
  Clk = 0;
  Enable = 0;
  Load = 0;
  Reset = 1;
  #10;
  Clk = 1;
  #10;
  Clk = 0;
  #10;
  Clk = 1;

 #10;
  Clk = 0;
   #10;
  Clk = 1;
  #10;
  Clk = 0;
  #10;
  Clk = 1;
  $display("Test completed");

end

endmodule
```